Research Article

Jingyi Huang and Ming Liu*

Between Nationalism and Globalism: Framing Climate Change Risks in Chinese and American Newspapers

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Abstract: This study gives a corpus-assisted discourse study of the framing of climate risks in *China Daily* (CD) and *The New York Times* (NYT) to reveal the dynamic relations between risk framing, ideologies, and national interests. The findings show that the risk frame is more favored in NYT. While CD shows more consensus on the risks, NYT emphasizes the disputes over the risks. CD prefers to highlight risks to people and adopt a global frame to underline efforts to address the risks. However, NYT features a national frame and emphasizes the potentially negative impacts on the economy and finance. It tends to show the pessimistic views in facing the threats and thus legitimize the inactions of the US government in the climate control. Their divergences in framing climate risks are further explained in terms of the different functions of the two newspapers and the national interests of the two countries.

Keywords: climate change; risk; corpus-assisted discourse study; framing; media

1 Introduction

In an increasingly globalizing world, human beings are confronted with growing risks and potential harms, especially with more challenges from technological

***Corresponding author: Ming Liu**, PhD, Assistant Professor, Department of Chinese and Bilingual Studies, The Hong Kong Polytechnic University, Hung Hom, Kowloon, Hong Kong,
E-mail: ming1.liu@polyu.edu.hk
Jingyi Huang, PhD, Assistant Professor, School of Foreign Languages, Shenzhen Technology University, 3002 Lantian Road, Pingshan District, Shenzhen, Guangdong, China,
E-mail: huangjingyi@sztu.edu.cn

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disasters, political and economic unrest as well as infectious diseases (Beck 1999). Risk has thus become a grave concern in the present-day world (Boholm 2016) and people increasingly use it to understand certain social problems (Battistelli and Galantino 2019). However, risk is related to not only scientific calculation of the likelihood of an unfavorable event (Haimes 2009); but more importantly, to the communication and perception of uncertain features and potential harms (Battistelli and Galantino 2019). It is thus discursively constructed (Ushchyna 2018). Particular ways of communicating risks can influence people's understanding of political events and social issues. Therefore, news media increasingly report risks to conceptualize certain social issues, such as some controversial health and environmental issues (e.g., Borah et al. 2024; Chan 2016).

Climate change is one of the most pervasive environmental risks to human society in the 21st century (Liu and Huang 2022). To meet the challenge, immediate global attention and coordination are urgently needed (Han, Sun, and Lu 2017). Nevertheless, in recent years, international climate negotiations have been accompanied by controversies and gridlock (Painter et al. 2023). This can be explained by the divergent attitudes in different countries towards the risks of climate change due to their different national interests. As two largest carbon emitters in the world, China and the US are expected to take more responsibilities for addressing climate risks; especially at recent climate conferences, China and the US have become the focus of climate negotiations (Pan, Opgenhaffen, and Van Gorp 2019). However, the US, as a leading developed country, has been shying away from taking the responsibilities for coping with climate change and hesitant to join the international efforts to curb climate change (Merry and Mattingly 2023). By contrast, as the largest developing and a big rising country, China has taken a proactive stance towards climate change in recent years. It views climate change as a good opportunity to expand its global influence in international affairs and improve its international image (Chen and Liu 2024), through actively participating in international climate talks and facilitating global collaboration in carbon emission reductions (Fu and Wang 2022).

Since news media play a paramount role in shaping public understanding of a global issue of which they have no prior knowledge and beyond their immediate concern (Gillings and Dayrell 2023), it is thus of great interest and significance to examine how the news media from the two superpowers, China and the US, compete in the international arena to construct the issue and legitimize their national climate actions. Although there has been growing interest in examining the discursive constructions of climate change in the two countries (e.g., Pan, Opgenhaffen, and Van Gorp 2019; Song et al. 2021), most of these studies focus on their preferential ways of framing the issue as regards the (un)certainty of climate change (i.e., whether climate change is certain or uncertain) (e.g., Bailey, Giangola,

and Boykoff 2014) as well as the attribution of responsibilities (i.e., who are responsible for climate change) (e.g., Liang et al. 2014). Still, news framing of climate risks or potential impacts of climate change in the two countries needs more exploration. As emphasized by Liu and Lin (2021), risk framing has become an important discursive strategy in covering social issues because it can be used by news media to evoke public fears and offer the rationale for political actions. This indicates that news framing of risks is not only related to evidence-based scientific knowledge but also associated with politics and national interests. Therefore, this study examines the risk framing of climate change in two leading English-language newspapers in China and the US, namely *China Daily* (CD) and *The New York Times* (NYT). It tries to answer three research questions: (1) What are the preferential ways of framing climate risks in the two newspapers? (2) How do they reveal their different ideologies towards climate change? and (3) What are the socio-political factors behind their particular ways of risk framing?

2 Media Framing of Climate Risks

News media are essential in disseminating climate knowledge to the public and raising their awareness to minimize climate risks (Gillings and Dayrell 2023). However, climate knowledge is not simply related to "truths" or "facts". It is transmitted by the media in specific socio-political contexts (Gillings and Dayrell 2023). Influenced by different ideological viewpoints and political interests, news reports can frame climate risks in a particular way, i.e., to "promote a particular problem definition, causal interpretation, moral evaluation, and/or treatment recommendations" of the risks through selecting certain aspects of reality and making them more salient (Entman 1993, 52). It thus has the potential to shape people's perception of climate risks and legitimize policies to address climate change (Vu, Liu, and Tran 2019). In recent decades, media representations of the potential impacts of climate change have attracted growing research interest (e.g., Painter 2013; Painter et al. 2023). For instance, Painter (2013) makes a distinction between the implicit/disaster and the explicit risk frames. He proposes that the meaning of risks can involve "a broad sense of a possible adverse impact" and "a more technical sense of assigning probabilities or confidence levels to different outcomes" (Painter 2013, viii). The implicit risk frame refers to the possible adverse impacts of climate change, which is often associated with the disaster frame. In this fear-based disaster frame, climate change is perceived as uncontrollable and unprecedented, which is usually not accompanied by positive messages. It is thus effective to attract media attention but ineffective to improve public engagement. The explicit risk frame refers to the chance that something negative will happen, which is associated with the use of the word

"risk" and the related concepts in the same semantic field. This frame appears to be more effective to communicate climate change than strong messages of disaster. However, it is found that the implicit/disaster risk frame is still predominantly used by journalists to highlight the serious outcomes of the changing climate (Merry and Mattingly 2023; Painter 2013).

The choice of different risk frames can communicate different ideologies and suggest different political interests. This can be witnessed in the analysis of the UK and US media. For example, Sonnett (2022) finds that political preferences can possibly influence the way of framing climate risks in the US media. Left-leaning newspapers prefer to link climate change to the word *risk*, which emphasizes the decision and choice. By contrast, the right-leaning media tend to associate global warming with the word *danger*, which foregrounds the passive role of humans in managing the risk. In the UK context, left-wing newspapers are found to emphasize the risks of global warming and create a sense of danger, whereas right-wing newspapers tend to question the dangers of global warming (Carvalho 2007). Nevertheless, these studies mainly investigate newspapers in a particular country. Few studies explore the framing of climate risks across countries and illuminate the influence of the national interests (exceptions including Grundmann and Krishnamurthy 2010; Schmidt, Ivanova, and Schäfer 2013; Shehata and Hopmann 2012). Furthermore, previous studies mostly rely on manual content analysis (e.g., Dirikx and Gelders 2010; Painter et al. 2023) rather than corpus-assisted discourse analysis (Liu and Li 2017). While manual content analysis contributes to identifying particular ways of risk framing at the macro-level of texts, it usually fails to address the discursive strategies and linguistic features that contribute to risk framing at the meso- and micro-levels of texts. Besides, the manual content analysis of a large number of texts is also time-consuming and labor-intensive, and the accuracy and replicability of analysis are often questioned. In order to remedy the above problems, this study proposes a corpus-assisted discourse study (CADS) approach to the analysis of the particular ways of framing climate risks in news media from different socio-political contexts.

3 Corpus-Assisted Discourse Studies

This study intends to combine the theories of framing (Entman 1993), critical discourse analysis (CDA) (Wodak and Meyer 2016), and corpus linguistics (CL) (Baker 2006), and gives a CADS of media framing of climate risks in both CD and NYT. CDA regards discourse as a social practice and takes a particular interest in uncovering the hidden power and ideologies behind the discourse (Fairclough 1992; Wodak and Meyer 2016). An ideology refers to "a worldview, a system composed of related

mental representations, convictions, opinions, attitudes, values and evaluations" shared by certain social groups (Reisigl and Wodak 2016, 25). Since news media serve as the main site of ideological struggles (Fairclough 1995), CDA has been widely adopted to unmask the ideological standpoints in the newspapers. However, the traditional qualitative analysis in CDA is often criticized for cherry-picking data or linguistic features for analysis (Baker and Levon 2015). Therefore, a growing number of studies have adopted CL methods in CDA (Alexander 2010; Gillings and Dayrell 2023), which are known for the potential to efficiently process a large amount of data, identify linguistic patterns with corpus analytic tools, and provide the basis for further detailed examination (Baker 2006; Islentyeva 2020).

CADS combines both quantitative and qualitative analyses to provide a deeper understanding of the discourse. The quantitative analysis relies on corpus analytic tools to examine a corpus in its entirety (Gillings, Mautner, and Baker 2023). These tools help to identify linguistic patterns and statistical trends which cannot be detected by manual analysis and reduce subjective involvement in the examination. While corpus analytic tools are effective at revealing recurring patterns, qualitative interpretation is necessary to explore the meaning behind these patterns. In CADS, the qualitative analysis can help explain and interpret the patterns and strategies shown by corpus analytic techniques from the critical perspective and expose the underlying ideologies (Baker et al. 2008; Partington, Duguid, and Taylor 2013). In this sense, CADS underlines a synergy of the methods related to CL and theoretical underpinnings related to CDA. According to Entman (1993, 52), framing "essentially involves selection and salience" and can be "manifested by the presence and absence of certain keywords". There has been growing literature using framing theory to examine media constructions of climate risks. However, previous studies mostly explore the presence of frames in the discourse and rely on the quantitative content analysis (e.g., Painter 2013; Vu, Liu, and Tran 2019). The detailed linguistic means of framing climate risks as well as the more nuanced ideological meanings within the dominant framing, by contrast, remain under-researched (Hart 2023). Given that climate risks are discursively constructed in nature, a CADS approach to framing analysis can combine the quantitative analysis of a large amount of data with the qualitative analysis of some discursive patterns and strategies which are key to news framing and make the analytic procedure more replicable (Alexander 2010; Liu and Li 2017; Wang and Ma 2021).

4 Data and Methodology

Two corpora have been built in this study, namely the CD corpus and the NYT corpus. The CD corpus is composed of news reports related to climate change in *China*

Daily (CD). The NYT corpus consists of news reports concerning climate change in *The New York Times* (NYT). The two newspapers were selected because they are important in expressing their respective countries' national perspectives and safeguarding their national interests on the global stage (Lee et al. 2002). NYT is considered as "newspaper of record" in the US (Liu and Huang 2022, 4), and is known for its agenda-setting influences on international news media (Song et al. 2021). It is often tracked by politicians for decision-making because it provides a window into public opinion (Song et al. 2021). As the largest and the most influential English-language newspaper in the Chinese mainland, CD serves as the party organ and shoulders the responsibilities for communicating the voice of the Chinese government to the external world (Liu 2023).

The data were retrieved from the online database LexisNexis and all the news reports containing *climate change* in the headlines from 2001 to 2020 were collected. The period from 2001 to 2020 captures significant developments in climate discourse, including the US' withdrawal from the Kyoto Protocol and the Paris Agreement, and the increasing global visibility of climate change impacts. To make sure that all the collected data were relevant to the topic, the downloaded reports were further manually checked. In general, 438 news reports were collected in the CD corpus, with 291,958 tokens. 1071 newspapers were collected in the NYT corpus, with 1,106,660 tokens. It shows that CD pays less attention to presenting climate change, which can be explained by the fact that the Chinese government did not highlight climate issues in the national agenda until 2007 (Broadbent et al. 2016).

This study starts with an examination of the keywords related to danger and risk in CD and NYT by an online corpus analytic tool, Wmatrix 5.0. This software enables an automatic semantic analysis and is distinctive because the UCREL Semantic Analysis System (USAS) it incorporates can categorize the English categories into 21 semantic fields, which can be further grouped into 232 semantic categories (SMCs) (Rayson 2008). Wmatrix 5.0 can generate a key SMC list by comparing the SMCs of a purpose-built corpus with those of a specialized or general reference corpus. The key SMCs are ranked in terms of their log-likelihood (LL) values. The higher their LL values, the more statistically significant those SMCs. Two key SMCs lists are generated by comparing the two corpora respectively with a general reference corpus, the 5-million-sample corpus of the British National Corpus (BNC). This study first focuses on the top 15 key SMCs in each corpus to examine the prominent topics/themes in each corpus. Then this study focuses on the SMC "Danger" (**A15-**) and investigates the actual use of the top 10 frequent tokens in the two newspapers to examine the prominence of the explicit and implicit/disaster risk frames. It is followed by a close analysis of the most frequent tokens (i.e., *risk*[*s*]) in **A15-**. To further examine their particular ways of risk framing in the two corpora, this study focuses on three aspects of risk framing: (1) the (un)certainty of climate

risks; (2) the causes and consequences of climate risks; and (3) the solutions/actions to climate risks (Entman 1993; Liu and Li 2017). Different discursive strategies are further examined to reveal how they can serve ideological purposes.

5 Findings

5.1 Analysis of the Key SMCs

Table 1 demonstrates the top 15 key SMCs in two corpora to identify the most prominent features and major trends in the two corpora. The shared 11 key SMCs indicate that the two newspapers have many similarities in representing climate change. An investigation of the most frequent tokens in each SMC suggests that the news coverage is closely related to four topics, including (1) environment; (2) causes and impacts; (3) science; and (4) politics. The most prominent theme is "environment", including weather events: **W4** (e.g., *climate*, *weather*, *floods*, *droughts* and *storms*), climate change: **A2.1+** (e.g., *change*) and **O4.6** (e.g., *temperatures*), as well as environment: **W5** (e.g., *environmental* and *environment*), **W1** (e.g., *planet*) and **W3** (e.g., *earth*). Besides, causes and consequences of climate change are also highlighted, as can be seen from **O1.3** (e.g., *gas* and *carbon dioxide*) and **F4** (e.g., *agricultural*, *agriculture* and *crops*). Furthermore, both newspapers highlight the important roles of scientific communities (e.g., *scientists*, *science*, and *scientific* in **Y1**) and government (e.g., *government* and *nations* in **G1.1)** in climate change representations.

Nevertheless, different key SMCs suggest their preferential ways of framing the issue. CD prefers to frame climate change as an international issue and focuses on international cooperation in addressing the issue. This can be seen from CD's preference for **M7** (e.g., *countries* and *developing countries*), **Z2** (*China* and *United States*), and **S1.1.3+** (e.g., *conference*, and *meeting*). This is consistent with the finding of previous studies that Chinese official newspapers prefer to frame climate change from the perspective of global collective efforts (Song et al. 2021; Wu 2009). However, they cannot be found in the top 15 key SMCs of NYT, which can be attributed to NYT's preference for framing climate change as a domestic issue (Grundmann and Krishnamurthy 2010).

Besides, NYT puts more emphasis on the dangers, as can be seen from the prominence of **A15-** (e.g., *risk*, *risks*, and *danger*) in NYT. This tendency is also revealed by other key SMCs (LL>6.63) in NYT, such as "dead" (**L1-**), "worry" (**E6-**), "fear/shock" (**E5-**), and "damaging and destroying" (**A1.1.2**).

To further examine how climate risks are framed in the two newspapers, this study focuses on the most frequent tokens in **A15-**. The top 10 frequent tokens in **A15-** in CD and NYT are as follows.

Table 1: Top 15 key SMCs in the two corpora.

Rank	CD			NYT		
	Tagset	**LL**	**Semantic Category**	**Tagset**	**LL**	**Semantic Category**
1	**W4**	11,600.08	Weather	**W4**	15,058.69	Weather
2	**A2.1**+	6,582.65	Change	**A2.1**+	5,027.9	Change
3	**W5**	2,673.52	Green issues	**W5**	3,268.13	Green issues
4	M7	1,740.8	Places	**Y1**	3,174.18	Science and technology in general
5	**W3**	1,582.52	Geographical terms	**W3**	2,777.2	Geographical terms
6	**Y1**	1,537.86	Science and technology in general	Z99	2,705.93	Unmatched
7	Z2	1,513.79	Geographical names	O4.6+	2,641.42	Temperature: Hot/on fire
8	**O1.3**	1,508.16	Substances and materials: Gas	**O1.3**	2,571.9	Substances and materials: Gas
9	**G1.1**	1,319.41	Government	**Q2.1**	2,563.52	Speech: Communicative
10	**F4**	1,310.69	Farming & Horticulture	**G1.1**	2,078.43	Government
11	**W1**	1,207.02	The Universe	**W1**	1,470.15	The Universe
12	X5.2+	1,129.36	Interested/excited/energetic	**O4.6**	1,011.38	Temperature
13	**Q2.1**	1,066.53	Speech: Communicative	A13	865.92	Degree
14	S1.1.3+	966.58	Participating	**F4**	860.84	Farming & Horticulture
15	**O4.6**	749.67	Temperature	A15-	756.8	Danger

CD: *risk* (123), *risks* (68), *danger* (17), *dangerous* (15), *dangers* (10), *exposure* (10), *hazards* (8), *peril* (4), *perils* (3), *jeopardy* (2)
NYT: *risk* (555), *risks* (414), *dangerous* (142), *dangers* (84), *danger* (75), *endangered* (66), *exposure* (32), *hazards* (31), *risky* (30), *hazard* (22)

These tokens can be semantically grouped into tokens expressing the meaning of risks and dangers (Boholm 2012). It suggests that the two newspapers prefer to use both explicit and implicit/disaster risk frames (Painter 2013). Tokens concerning risks are *risk* and *risks* in CD (191, 76.4 %) and *risk*, *risks* and *risky* in NYT (999, 70.4 %). Tokens concerning dangers are *danger*, *dangerous*, *dangers*, *hazards*, *peril*, *perils* and *jeopardy* in CD (59, 23.6 %) and *dangerous*, *dangers*, *danger*, *endangered*, *hazards* and *hazard* in NYT (420, 29.6 %). The percentages of the explicit and implicit/disaster risk frames suggest that while the explicit risk frame is more favored in both corpora than the implicit/disaster frame, NYT shows more preference for the implicit/disaster risk frame than CD. Figures 1 and 2 show the diachronic distribution of the implicit and explicit risk frames. It shows that while the explicit risk frame rises in both corpora, the implicit/disaster frame rises in CD but decreases in NYT. The growing use of the implicit/disaster frame in CD in

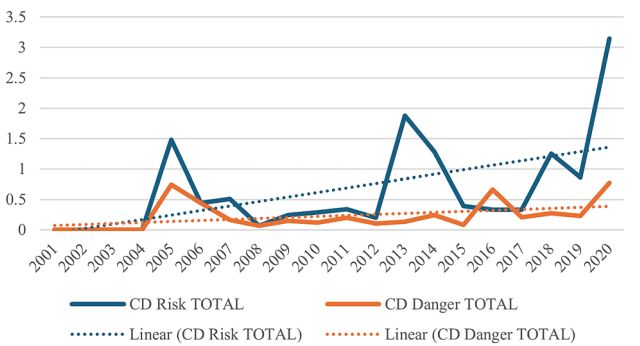


Figure 1: Standardized frequencies of the tokens in each year in CD.

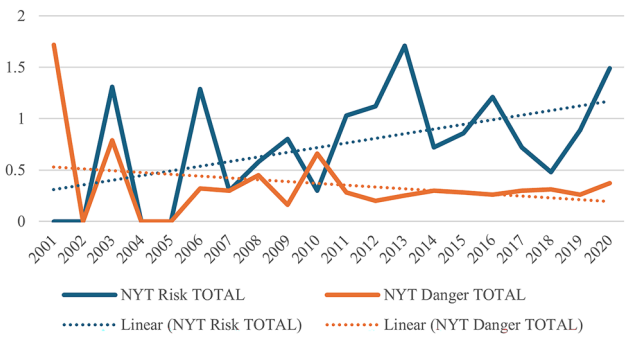


Figure 2: Standardized frequencies of the tokens in each year in NYT.

recent years reflects its emphasis on the consequences, underscoring the urgency of addressing the issue. Conversely, NYT's increasing reliance on the explicit/risk frame suggests its tendency to highlight uncertainty surrounding climate impacts. However, despite its decline, the implicit/disaster frame remains relatively prominent in NYT. This suggests a continued reliance on fear-based language to frame climate change, which can draw more media attention but also risk creating despair and leading to public apathy (Painter 2013).

5.2 Analysis of *Risk(s)*

Since *risk(s)* are the most frequent tokens used in **A15-**, a close analysis of the tokens *risk(s)* can help examine the particular ways of risk framing in CD and NYT. An examination of the concordance lines of *risk(s)* finds that 169 instances in CD (85.4 %) and 728 instances in NYT (87.1 %) are related to climate risks, suggesting both corpora's preference for emphasizing climate risks. Among them, *risk(s)* are predominantly used as nominal forms in the two corpora, with 165 instances in CD (97.6 %) and 723 instances in NYT (99.3 %). The predominant use of the nominal forms in the media can help to highlight the present moment of the risks and suppress the future aspects. It can also contribute to describing the risks as presupposed and assumed (Dunmire 2005). The following sections give a close examination of these tokens in the specific contexts to examine how they contribute to the framing of risks in terms of (1) the (un)certainty of the risks; (2) the causes and consequences of the risks; and (3) the solutions/actions to reduce the risks.

5.2.1 The (Un)certainty of Risks

A close reading of the concordance lines in CD shows that 166 (98.2 %) instances highlight the certainty of the risks and only 3 (1.8 %) instances suggest the uncertainty. Therefore, CD prefers to underline the certainty of climate risks. In NYT, 625 instances express the certainty of the risks (85.9 %), while 103 instances suggest the uncertainty (14.1 %). Therefore, NYT shows more preference for the uncertainty of climate risks than CD.

The certainty of climate risks is constructed through several linguistic means in both corpora. Firstly, explicit assertions are often used to make the existence of risks uncontested (Liu and Zhang 2018), with 86 (51.8 %) instances in CD and 214 (34.2 %) instances in NYT, as in the following:

(1) According to estimates, climate change *risks* **will threaten** as much as 30 percent of global GDP by 2025, double the amount in 2008. (*CD*, July 20, 2020)

Secondly, the certainty of climate risks is often implied by those verbs or verb phrases that occur with *risk(s)*, such as *disclose**, *warn**, *know**, *recognize**, *highlight**, *underscore**, *accept**, *understand**, *take…account*, *believe**, *acknowledge**, *emphasize**, *find**, etc. These verbs or verb phrases usually carry the implicature that what is followed is true. Examples are as follows:

(2) Many uncertainties remain, but we know enough to **recognize** that there are large long-term *risks.* (*CD*, November 28, 2007)

(3) He **acknowledged** that the *risks* of climate change comprise the bulk of the report, but there are also "really exciting" opportunities. (*NYT*, December 13, 2020)

Besides, the acceptance of the risks is also demonstrated through predication strategies, i.e., the ways in which social events are qualified (Reisigl and Wodak 2016), as witnessed by the adjectives that co-occur with *risk(s)*, such as *real*, *existential*, and *existing*, *known*, *ubiquitous*, *visible*, *actual*, and *true*. Examples are as follows:

(4) We do still have time to prevent the worst effects of climate change, but the world must do much, much more to deal with this **existential** *risk.* (*CD*, October 10, 2013)

(5) "The *risks* are **real**," he said, "and those of us who live here in South Florida observe them on a daily basis." (*NYT*, July 21, 2020)

While NYT accepts that climatic issues can lead to future risks, the explicit causal links to the potential impacts are sometimes hedged by verbal process verbs (Halliday 1994) such as *discuss*, *comment*, *communicate*, *talk about*, *debate*, etc., and mental process verbs such as *assess**, *evaluate*, *study*, *gauge*, *monitor* and *predict*. They suggest that the risks can be argued. For example:

(6) "**Commenting on** climate *risks* for purely partisan reasons would be a mistake," he said. (*NYT*, November 08, 2019)

(7) This month, the Fed **assessed** climate change *risks* in detail in a financial stability report, a first for the twice-yearly document. (*NYT*, November 08, 2020)

The uncertainty of the risks is also revealed by highlighting the unwillingness of some businesses to disclose the risks (25) and their misrepresentation of the risks (15). It thus helps to downplay the certainty of the risks, as in the following:

(8) **Just a fraction** of global businesses currently **disclose** the financial *risks* posed by climate change. (*NYT*, November 23, 2020)

(9) New York's attorney general sued Exxon Mobil on Wednesday, claiming the company defrauded shareholders by **downplaying** the expected *risks* of climate change to its business. (*NYT*, October 24, 2018)

In addition, NYT tends to construct the uncertainty of the risks by referring to doubts about the risks (17). However, the explicit negative stance of the newspaper can be shown by the use of concession (Martin and White 2005), as in the following:

(10) As Exxon Mobil responded to news reports in 2015 that said that the company had spread doubt about the *risks* of climate change **despite** its own extensive research in the field… (*NYT*, August 23, 2017)

5.2.2 Causes and Consequences of Risks

A close examination finds that 155 instances in CD (71.8 %) and 543 instances in NYT (60.7 %) suggest the causes and consequences of the risks. According to Fillmore and Atkins (1992), the source of threat and risk object are two important constituents in framing risks (see also Ushchyna 2018). The source of threat refers to the causes of the risks. Risk object refers to the consequences of the risks. As in the following example, rising sea levels and extreme weather events are constructed as the causes, and heavily populated coast is considered as the consequence.

(11) Rising sea levels and more extreme weather events (causes) such as Typhoon Mangkhut also put its heavily populated coast (consequence) at *risk*. (*CD*, December 13, 2018)

A close examination of the causes and consequences of climate risks finds that different classification strategies are used (van Leeuwen 2008). In both corpora, the risks mainly originate from climate change/global warming (72.6 % in CD and 64.4 % in NYT) and the related natural events (27.4 % in CD and 35.6 % in NYT), such as *flood*, *extreme weather events*, *heat waves*, *droughts* and *wildfires*. However, natural disasters are more frequently mentioned in NYT (150, 35.6 %). It thus helps to background the human causes of the risks.

Furthermore, the causes of the risks are also dramatized in NYT. It is revealed through predication strategies, as can be seen from such descriptions to modify the source of threat as *rising seas and* ***heavier***, ***more frequent*** *rain*, ***mega****droughts,* ***long*** *dry periods*, ***large***, ***destructive*** *wildfires*, ***catastrophic*** *fire*, ***more frequent*** *and* ***severe*** *droughts*, ***increasingly severe*** *storms*, ***powerful*** *storms*, ***more intense*** *hurricanes, and the increased occurrence of* ***extremely hot*** *days*, etc. These expressions all underline the severity of the events that can cause the risks.

Besides, the seriousness is also substantiated by adjectives collocated with *risk(s)* such as ***high****/risk* (34), ***increased****/risk* (15), ***greater****/risk* (13), ***significant****/risk* (10), ***growing****/risks* (8), ***significant****/risk* (8), ***increasing****/risk* (7), ***significant****/risks* (6), ***great****/risks* (6), ***higher****/risks* (6), ***greatest****/risks* (6), ***important****/risks* (5), ***serious****/risks* (5), ***severe****/risk* (4), ***critical****/risk* (4), ***catastrophic****/risk* (4), ***profound****/risks* (3), and ***major****/risks* (3). For example:

(12) Climate change is severely straining the world's oceans, creating **profound** *risks* for coastal cities and food supplies, a U.N. report finds. (*NYT*, September 27, 2019)

As for the consequences of the risks, CD and NYT show preferences for highlighting both objects and places at risk. In CD, four main objects at risk can be identified, namely "society and people" (44, 45.8 %), "business/economy" (27, 28.1 %), "animals/plants" (13, 13.5 %), and "political security" (12, 12.5 %). Among them, social threats are highly foregrounded in the media, especially concerning health, hunger and poverty. This indicates CD's preference for emphasizing the impacts on humans. It is consistent with the finding that Chinese media tend to foreground human interests (Xie 2015). For example,

(13) Those who have most at stake at the Paris conference are the 3.5 billion poorest people. They are the least able to cope with increased *risk* of ***floods, droughts, hunger and disease***, and are also least responsible for the emissions that have caused the problem. (*CD*, November 27, 2015)

The risks to humans are also shown in the emphasis on the dangers posed to food production. Intensifying expressions, such as "dramatically" in the following example, can be found to highlight the serious dangers to the food supply, which can have impacts on human development. For example,

(14) Climate change will put pressure on water sources and increase the *risk* of **droughts** in China, which in turn will dramatically affect crops such as rice. (*CD*, December 13, 2018)

NYT also highlights four aspects of the threats, including "threats to businesses, finance and economy" (145, 56 %), "social threats" (57, 22 %), "threats to animals and plants" (46, 17.8 %) and "threats to political security" (11, 4.2 %). Among them, "threats to businesses, finance and economy" takes the largest share. Specifically, the newspaper underlines threats to the "financial system" (28), "companies" (24), "facilities" (15), and "homes/home values" (14). It thus shows that NYT rarely links climate risks to humans. Instead, it prefers to relate risks to the economy and market. Meanwhile, predication strategies are frequently used to accentuate the economic risks, as can be seen by "greater" in Example 15.

(15) It could also put *roads, bridges and other infrastructure* at **greater** *risk*, he said, because developers would not be required, for instance, to analyze whether sea-level rise threatened to eventually submerge a project. (*NYT*, January 03, 2020)

The emphasis on economic loss is also reflected by the use of the quantification strategies (Baker 2006; Islentyeva 2020). Specific numbers can serve to make the financial impacts undebatable and foreground the high costs to companies, buildings and facilities, as in the following:

(16) The board estimates that there are significant climate *risks* for most companies it tracks, which **represent $27.5 trillion, or 93 percent**, of United States stocks as measured by market value. (*NYT*, September 27, 2016)

As regards places at risk, CD mainly emphasizes the dangers posed to "the world" (25, 40.3 %), "China" (16, 25.8 %), and "less developed areas" (9, 14.5 %). It thus further supports the above finding that CD prefers to frame climate change as a global issue/risk. Hereby it helps to encourage collective actions to fight against the risks. For example,

(17) Beyond this, even half a degree will significantly worsen the *risks* of drought, floods, extreme heat and poverty for hundreds of millions of people **around the world.** (*CD*, December 13, 2018)

Besides, it is interesting to find that local risks are occasionally backgrounded in CD, as evidenced by some mitigating expressions, such as *a **moderate** increase of risks*, *a **relatively** high increase*, etc. Positive implications of climate change can also be identified to make climate change less threatening to China. It corresponds with CD's preference for underlining the globalized feature of climate risks. For instance,

(18) As the report shows, while climate change **may well bring positive implications for some regions of the country**, overall climate change poses serious *risks* for sustaining development in the decades to come. (*CD*, April 20, 2007)

Finally, CD also discusses the risks to people living in less developed areas. Intensification strategies are used to highlight that they are at grave risk. These representations help to underline the need to support developing countries and indicate the responsibilities of developed countries for offering assistance, as in the following:

(19) **In developing countries**, vulnerable populations face the double jeopardy of being at greater *risk* and having limited capacity to reduce *risk* through private markets. (*CD*, April 20, 2007)

On the contrary, NYT mainly concerns the risks to "the US" (106, 60.2 %), "other countries or areas" (41, 23.3 %) and "the world" (29, 16.5 %). It suggests that NYT prefers to frame climate change as a domestic issue/risk. It is consistent with the view that the US news coverage of natural disasters is prone to be locally focused (Parks 2020). The geographical closeness of the risks can potentially dramatize climate risks, and

the frequent use of amplifying adjectives also contributes to accentuating the severe impacts on the US, such as *significant*, *major*, *important*, *growing*, *increased*, *higher*, *severe*, *intense*, etc., for example:

(20) Extreme precipitation is likely to increase with rising temperatures because of growing atmospheric humidity, leading to a **higher** *risk* of flash flooding **nationwide**. (*NYT*, December 06, 2016)

In some cases, numbers are also used to foreground the severity of climate risks to the US, highlighting the contribution of climatic events to the US' vulnerable populations, facilities and financial costs. As in the following, the use of "96,000" helps to frame climate change as uncontrollable and undefeatable, and thus evokes readers' fear.

(21) "In 2020, nearly **96,000 people in the Long Beach area** alone may be at *risk* from sea-level rise," the report said, referring to just one oceanfront community on the South Shore of Long Island. (*NYT*, November 17, 2011)

Besides, NYT also discusses the risks to coastal cities, low-lying areas and tropical countries (12), but impacts on poorer countries are backgrounded (3). It thus indicates the US' intention to evade responsibilities for helping poorer countries to address climate risks. For instance,

(22) **Low-lying, flood-prone areas** are at particularly high *risk* of becoming unlivable – or at least uninsurable. (*NYT*, January 20, 2019)

5.2.3 Actions to Address Risks

A close examination of the concordance lines reveals that 35 (16.2 %) instances in CD discuss actions to address the risks while only 76 (8.5 %) instances in NYT are concerned with solutions to the risks. It thus implies CD's inclination to show the responsibilities for reducing the risks, while NYT tends to shirk responsibilities.

An examination of the 36 concordance lines in CD finds that 25 examples explicitly refer to the actors taking the responsibilities, which mainly include "global communities" (12), "China" (5) and "developing countries" (3). They suggest that CD prefers to attribute responsibilities to the world. This is consistent with the finding of previous studies that party-sponsored Chinese mainstream newspapers prefer to use the collaboration frame in framing climate change (Han, Sun, and Lu 2017). As in the following example:

(23) We do still have time to prevent the worst effects of climate change, but **the world** must do much, much more to deal with this existential *risk*. (*CD*, October 10, 2013)

In addition to highlighting the collective actions to deal with the risks, CD also expresses optimistic attitudes towards the actions, as evidenced by some positive words, such as *effort(s)*, *progress*, *working to*, *effective*, etc. This can potentially help construct climate risks as manageable and climate actions as feasible. It thus underlines the proactive position of the Chinese government in coping with the risks. For instance,

(24) The Paris agreement proposes to keep the global mean temperature increase to well below 2 C above preindustrial levels, and to pursue **efforts** to limit warming to 1.5 C, to reduce the related *risk* and impacts. (*CD*, December 13, 2018)

An examination of the 74 concordance lines in NYT finds that 48 instances mention the actors to reduce the risks. In particular, the media mainly attribute responsibilities to its own country rather than foreign countries (42, 87.5 %). It can be explained by the US' lack of interest in highlighting international relations in reporting global issues (Xie 2015). Four main groups can be identified, namely "companies" (10), "government" (9), "the public" (7) and "cities" (4). It finds that NYT tends to shift responsibilities from the government to other groups to cope with the risks, notably the insurers, as in the following example:

(25) It is clear that California's **insurers** are struggling to prepare themselves for a new era of accelerating climate *risk*. (*NYT*, December 05, 2019)

In addition to foregrounding responsibilities for reducing the risks, NYT also describes unpromising attitudes towards the actions. This can be seen by the media's emphasis on costly plans and limited tools (14). These constructions can picture climate change as hard to control, which can potentially legitimize the government's inaction to cope with the risks. For example:

(26) But right now, we **don't have** enough information, and we **don't have the right financial products** to insure this *risk*. (*NYT*, June 11, 2019)

6 Discussion and Conclusions

To sum up, this study gives a corpus-assisted discourse study of the framing of climate risks in CD and NYT and reveals both differences and similarities in their framing of climate risks. Both corpora prefer to use the explicit risk frame but NYT shows more tendency to use the implicit/disaster risk frame. It thus indicates its potential to create fear and despair. Although the explicit risk frame is viewed as a more effective way to communicate climate change (Painter 2013), preferential ways of framing risks in the two newspapers indicate their different attitudes towards climate change. CD prefers to frame climate change as a global issue and

tends to underline the certainty of climate risks and the impacts on human beings and the world. Therefore, it pays attention to the collective efforts to manage the risks. NYT tends to frame climate change as a national issue and underline the uncertainty of climate risks. It prefers to highlight the impacts of climate change on business and the economy and shift the US government's responsibilities for addressing the issue.

Their preferential ways of framing climate risks can be explained in terms of the different functions of the two newspapers and the national interests of the two countries concerning climate change (Han, Sun, and Lu 2017; Song et al. 2021; Wu 2009). As an official English-language newspaper, CD serves as the mouthpiece of the Chinese government and shoulders the responsibility for communicating the voices of the Chinese government (Liu and Li 2017). With the release of the first national assessment of global warming report in 2006, the government built the political consensus on climate change and largely accepted the potential consequences of climate change, such as water shortage and food insecurity, on China's further development (He 2014; Wu 2009). In the last two decades, the Chinese government has played an increasingly active role in addressing climate change (Stensdal 2014). Therefore, CD prefers to "adhere closely to the Chinese government's guidelines to project one uniform voice in positioning China as the leader in the global battle against climate change" (Song et al. 2021, 18). The preference for a global frame and the emphasis on the certainty of climate risks underline the urgency and necessity of global efforts to address climate change and the positive role China plays in climate change issues. However, the theme of "international responsibility" may also indicate CD's tendency to place responsibilities in the hands of external agents, especially in developed countries (Liang et al. 2014; Xie 2015), since China tends to underline the differential responsibilities between developed and developing countries (Liu 2023).

Previous studies have demonstrated that countries with higher GDP per capita are inclined to frame climate change from the perspective of domestic politics rather than international relations (Song et al. 2021; Vu, Liu, and Tran 2019). This argument is also supported by NYT which frames climate risks as a contested issue and thus affects the public's and policymakers' perceptions of climate change. Besides, the journalism norm of the Anglo-American media system underlines the practice of balanced reporting (Hallin and Mancini 2004), so American newspapers tend to include different viewpoints on climate change (Boykoff and Boykoff 2004). This also contributes to representing the uncertainty of climate risks in NYT, even though some studies have revealed that there is a decrease in skeptical voices in US news representations of climate change (Xie 2015). Furthermore, previous studies have demonstrated that the US news media tend to align with national interests in

the coverage of international issues (Lee et al. 2002). This study supports the argument and finds that NYT follows the neoliberalism ideology in framing climate risks that emphasizes economic growth and free market logic and reduces government solutions. The frame of climate change as a domestic issue and the emphasis on the uncertainty of climate risks help to downplay the severity of climate change and shift the global responsibilities of the US for tackling climate change. It can thus help legitimize the US' failure to adopt an effective climate plan (Fremstad and Paul 2022). For example, although the Clinton and Obama administrations were actively involved in climate negotiations, the US gradually concluded its leadership in international climate politics and abdicated the global responsibilities in climate change actions after President Trump announced the withdrawal from the Paris Agreement in 2017 (Kopra 2019).

This study has several limitations. First, it focuses on two leading newspapers in China and the US. Future research could examine other party-oriented or market-oriented Chinese newspapers and US newspapers with varying partisan orientations. Second, while this study analyzes climate change coverage, future studies could explore news framing of related environmental issues, such as global warming and heat waves. Third, this study investigates news reports from 2001 to 2020. Future studies could include more recent data to uncover new trends in news framing, especially during and after the COVID-19 pandemic. It is expected that future studies can benefit from a corpus-assisted discourse study as it can reveal not only the preferential ways of framing but, more importantly, the hidden attitudes and ideological positions of different newspapers (Song et al. 2021). The use of CL methods can contribute to the identification of frames in a more scientific and replicable way, while the close analysis of some linguistic tokens in their specific contexts can help to reveal the detailed linguistic strategies used by different newspapers for the construction of news frames (Baker 2006). Therefore, a combination of the methods and theories of CL, CDA and framing can expose the news framing of risks at different levels. More importantly, it underlines the socio-political factors behind the particular ways of risk framing and argues for the examination of risk framing in its socio-political contexts (Ushchyna 2018).

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Bionotes

Jingyi Huang
PhD, Assistant Professor, School of Foreign Languages, Shenzhen Technology University,3002 Lantian Road, Pingshan District, Shenzhen, Guangdong, China
huangjingyi@sztu.edu.cn

Jingyi Huang is an Assistant Professor at the School of Foreign Languages, Shenzhen Technology University. Her research interests cover critical discourse analysis, corpus-assisted discourse studies, environmental discourse analysis and climate change discourse analysis.

Ming Liu
PhD, Assistant Professor, Department of Chinese and Bilingual Studies, The Hong Kong Polytechnic University, Hung Hom, Kowloon, Hong Kong
ming1.liu@polyu.edu.hk

Ming Liu is an Assistant Professor at the Department of Chinese and Bilingual Studies, The Hong Kong Polytechnic University. His research interests cover critical discourse analysis, corpus linguistics, pragmatics, systemic functional linguistics, and intercultural communication. His recent publications have appeared in some international journals, such as *Journalism*, *Discourse & Communication*, *Language & Communication*, *Journal of Language and Politics*, *Text & Talk*, *Discourse, Context & Media*, *Lingua* and *Critical Arts*.